Oleksandra Golub 856706

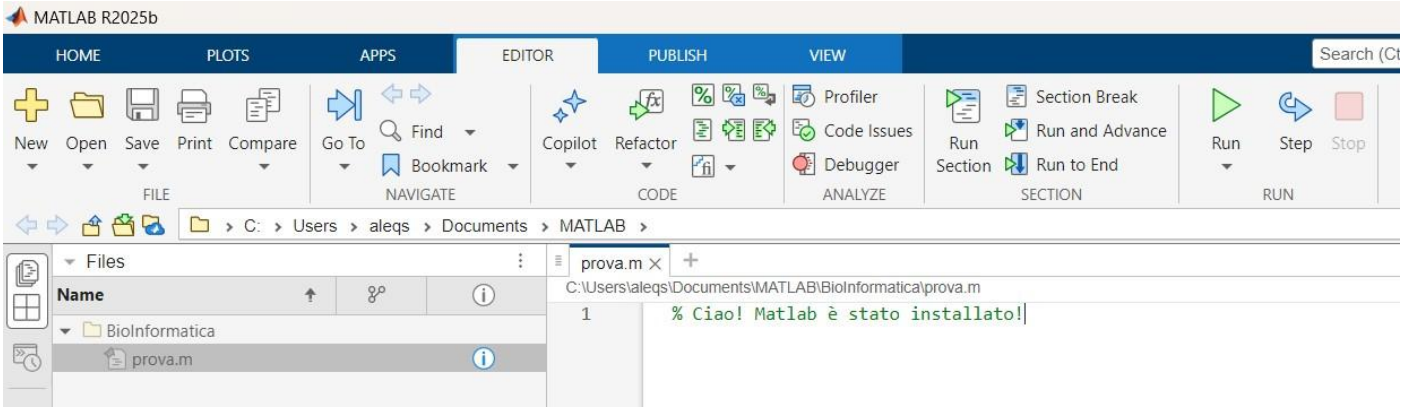
Hands On

Make sure that you have downloaded and installed on your computer at least Matlab, R (RStudio), Wolfram/Mathematica, Python and Julia.

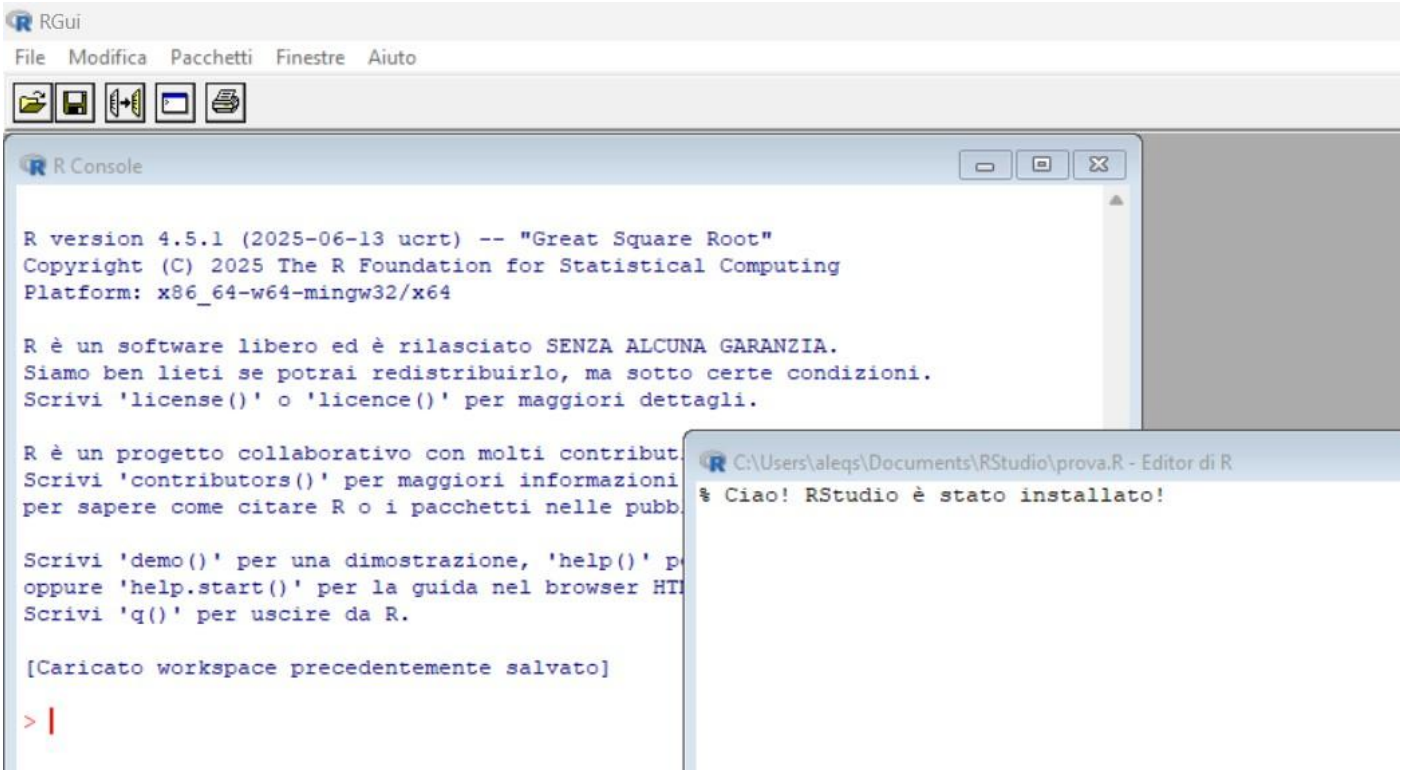
- For R, please install also **Bioconductor** (www.Bioconductor.org) – use the instructions on the site, i.e., the **`BiocManager::install(...)`** function
- Also check **BioPython** and **BioJulia**

Report (on Moodle) your progress by posting screenshots of the final installation

1) Matlab:



2) RStudio



3) RStudio con BioConductor

```
RGui - [R Console]
File  Modifica  Visualizza  Varie  Pacchetti  Finestre  Aiuto

Messaggio di avvertimento:
2 packages out-of-date; 0 packages too new
> BiocManager::version()
[1] '3.21'
> BiocManager::valid()

* sessionInfo()

R version 4.5.1 (2025-06-13 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 22631)

Matrix products: default
  LAPACK version 3.12.1

locale:
[1] LC_COLLATE=Italian_Italy.utf8  LC_CTYPE=Italian_Italy.utf8    LC_MONETARY=Italian_Italy.utf8 LC_NUMERIC=C
[5] LC_TIME=Italian_Italy.utf8

time zone: Europe/Rome
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

loaded via a namespace (and not attached):
[1] BiocManager_1.30.26 compiler_4.5.1      tools_4.5.1

Bioconductor version '3.21'

  * 2 packages out-of-date
  * 0 packages too new

create a valid installation with

  BiocManager::install(c(
    "BiocParallel", "Rsamtools"
  ), update = TRUE, ask = FALSE, force = TRUE)

more details: BiocManager::valid()$too_new, BiocManager::valid()$out_of_date

Messaggio di avvertimento:
2 packages out-of-date; 0 packages too new
>
```

4) P con BioPython

```
PS C:\Users\aleqs> pip install biopython
Collecting biopython
  Downloading biopython-1.85-cp313-cp313-win_amd64.whl.metadata (13 kB)
Requirement already satisfied: numpy in c:\users\aleqs\appdata\local\program
iopython) (2.1.2)
Downloading biopython-1.85-cp313-cp313-win_amd64.whl (2.8 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.8/2.8 MB 5.1 MB/s eta 0:00:00
Installing collected packages: biopython
Successfully installed biopython-1.85
```

```
PS C:\Users\aleqs\Desktop\Bioinformatica>  py prova.py
1.85
PS C:\Users\aleqs\Desktop\Bioinformatica>
```

5) Julia

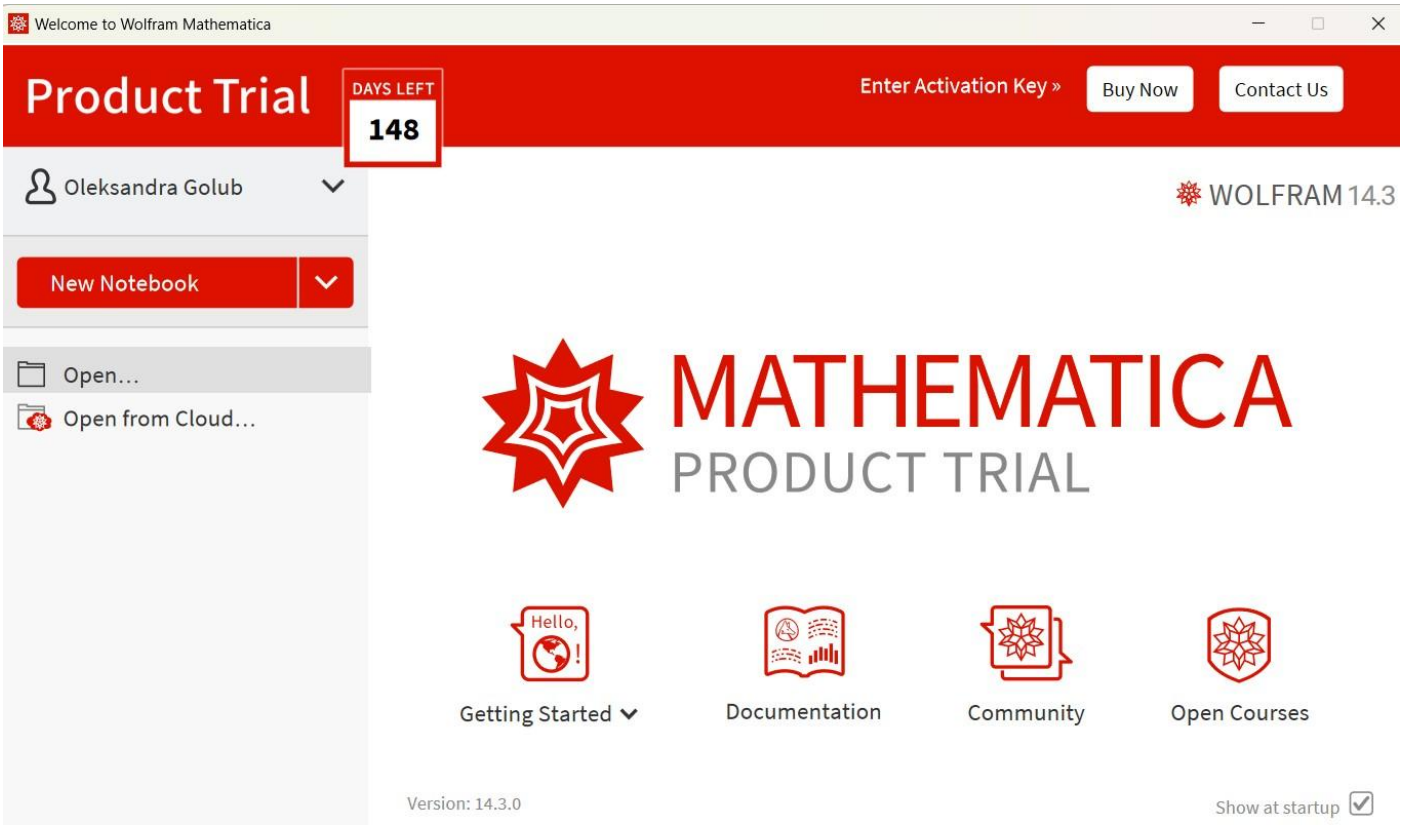
```
Julia 1.11.7

               _
   _       _ _(_)_     |  Documentation: https://docs.julialang.org
  (_)     | (_) (_)    |
   _ _   _| |_  __ _   |  Type "?" for help, "]?" for Pkg help.
  | | | | | | |/ _` |  |
  | | |_| | | | (_| |  |  Version 1.11.7 (2025-09-08)
 _/ |\__'_|_|_|\__'_|  |  Official https://julialang.org/ release
|__/                   |

julia> versioninfo()
Julia Version 1.11.7
Commit f2b3dbda30 (2025-09-08 12:10 UTC)
Build Info:
  Official https://julialang.org/ release
Platform Info:
  OS: Windows (x86_64-w64-mingw32)
  CPU: 8 × 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz
  WORD_SIZE: 64
  LLVM: libLLVM-16.0.6 (ORCJIT, tigerlake)
Threads: 1 default, 0 interactive, 1 GC (on 8 virtual cores)

julia>
```

6) Julia con BioJulia

```
(@v1.11) pkg> status
Status `C:\Users\aleqs\.julia\environments\v1.11\Project.toml`
  [00701ae9] BioAlignments v3.1.0
  [7de43156] BioMakie v0.4.4
  [7e6ae17a] BioSequences v3.5.0
  [de9282ab] BioStructures v4.7.0
  [3c28c6f8] BioSymbols v5.2.0
  [336ed68f] CSV v0.10.15
  [a93c6f00] DataFrames v1.8.0
  [31c24e10] Distributions v0.25.122
  [c2308a5c] FASTX v2.1.7
  [587475ba] Flux v0.16.5
  [899a7d2d] GenomicFeatures v3.0.0
  [add582a8] MLJ v0.21.0
  [91a5bcdd] Plots v1.41.1
```

7) Wolfram/Mathimatica